

Program-6 Docker Image Build and Push to Docker Hub and GitHub Container Registry

To build a Docker image for Program-6, run it locally, and push it to both **Docker Hub** and **GitHub Container Registry (GHCR)**. Also, to connect the local source code to GitHub.

2. Software Requirements

Component	Version
OS	Ubuntu / Linux
Docker	20.x or later
Docker Compose	1.29+
Git	Latest
GitHub Account	Required
Docker Hub Account	Required

3. Folder Structure

```
p6/  
├── Dockerfile  
├── README.md  
└── test.txt
```

4. Dockerfile

```
FROM alpine  
  
CMD ["echo", "Program 6 Docker build successful"]
```

5. Step-by-Step Procedure

Step 1: Navigate to Project Directory

```
cd ~/p6
```

Step 2: Build Docker Image

```
sudo docker build -t program-6 .
```

Step 3: Verify Image Creation

```
sudo docker images | grep program-6
```

Step 4: Run Docker Container

```
sudo docker run program-6
```

Expected Output:Program 6 Docker build successful

Step 5: Tag Image for Docker Hub

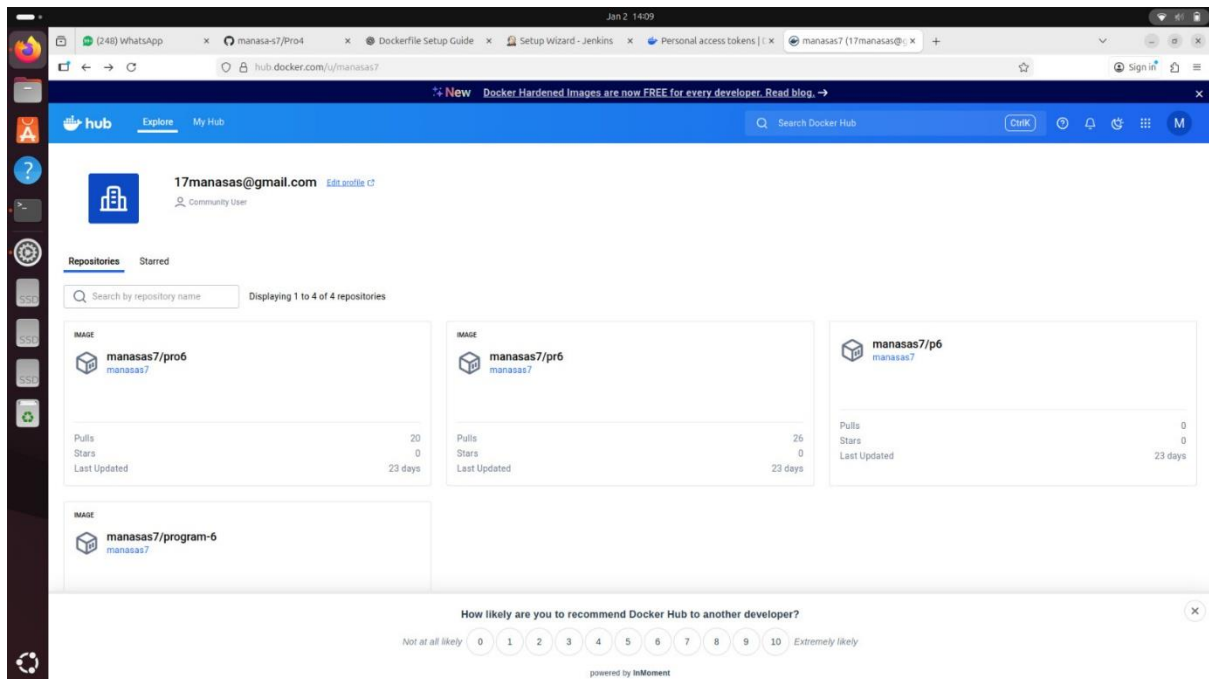
```
docker tag program-6 <dockerhub-username>/program-6:latest
```

Step 6: Login to Docker Hub

```
docker login
```

Step 7: Push Image to Docker Hub

```
docker push <dockerhub-username>/program-6:latest
```



Step 8: Tag Image for GitHub Container Registry

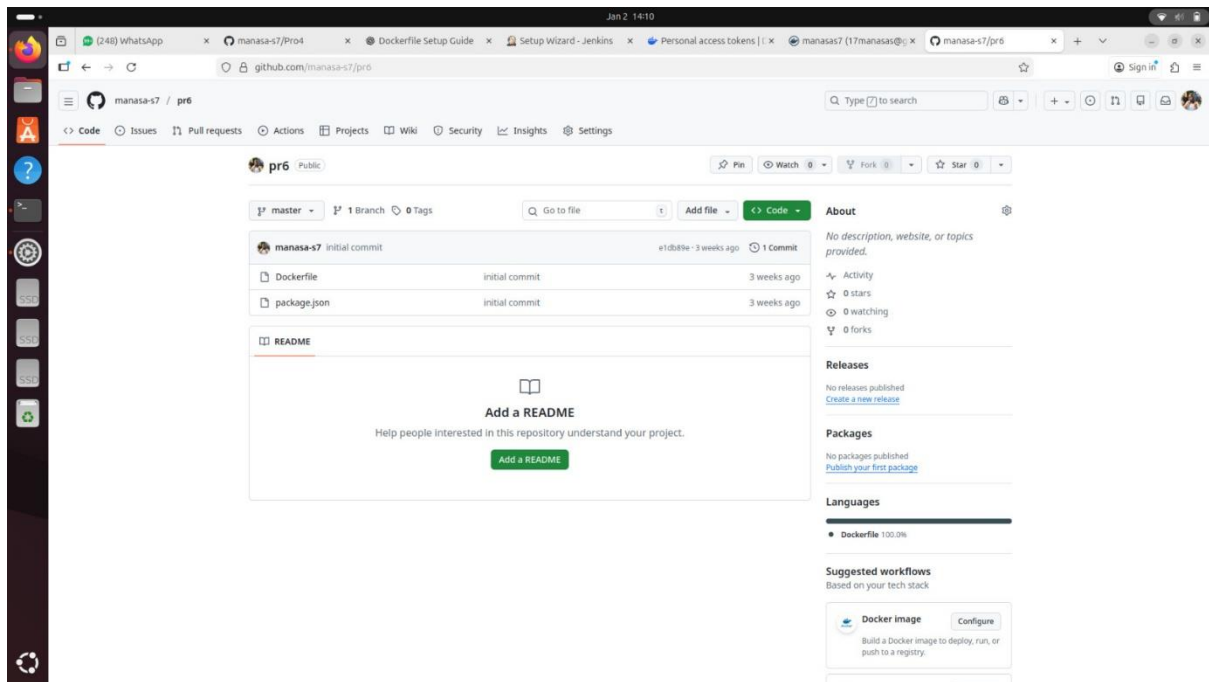
```
docker tag program-6 ghcr.io/<github-username>/program-6:latest
```

Step 9: Login to GHCR

```
echo <GITHUB_PAT> | docker login ghcr.io -u <github-username> --password-stdin
```

Step 10: Push Image to GHCR

```
docker push ghcr.io/<github-username>/program-6:latest
```



Step 11: Connect Local Code to GitHub

git init

git remote add origin https://github.com/<github-username>/program-6.git

git add .

git commit -m "Initial Program 6 commit"

git push -u origin main